

CHEMISTRY**Q1** Which of the following ion act as a strong oxidizing agent

- (1) Ce^{+3} (2) Gd^{+2} (3) Tb^{+4} (4) Eu^{+2}

Ans. (3)**Sol.** $\text{Tb}^{4+} + \text{e}^- \rightarrow \text{Tb}^{+3}$

Lanthanides prefer to exist in +3 oxidation state, so Tb^{4+} gain electron, hence behave as oxidizing agent.

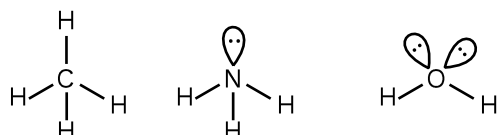
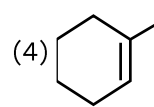
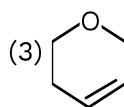
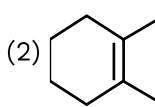
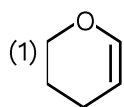
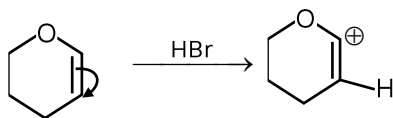
Q2 Difference of melting point and boiling point between O and S can be explained by

- (1) Electron gain enthalpy (ΔH_{eg}) (2) Electronegativity
(3) Atomic size (4) Atomicity

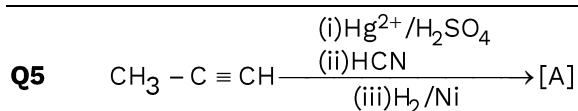
Ans. (4)**Sol.** Due to the more catenation property of S than O $\rightarrow$ M.P/B.P $\uparrow$ **Q3** State the correct statement for given compound H_2O , NH_3 and CH_4

- (a) All central atom are sp^3 hybridization
(b) Order of dipole moment is $\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$
(c) NH_3 and H_2O are basic in nature, NH_3 and H_2O are Bronsted-Lowry Acid and Bases respectively.
(d) Bond angle of H_2O , NH_3 and CH_4 respectively are 104.5° , 107° and 109.5° .

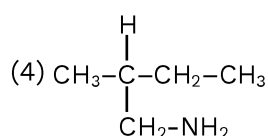
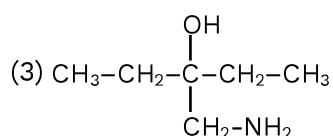
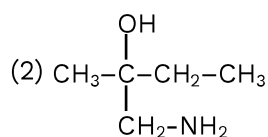
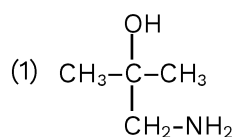
- (1) (a), (b) & (c) only (2) (a), (b) & (d) only
(3) (a) & (b) only (4) (a), (b), (c) & (d)

Ans. (2)**Sol.****Q4** Which compound react fastest with HBr**Sol.**

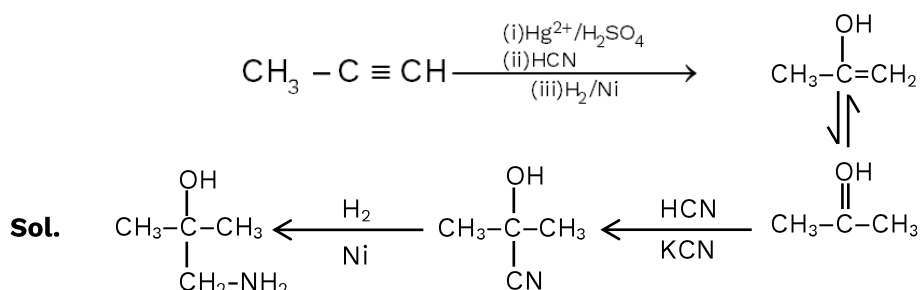
Stable due to resonance



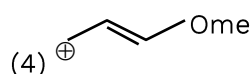
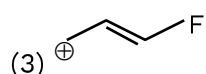
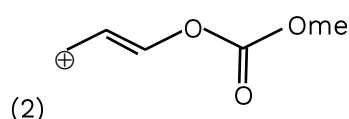
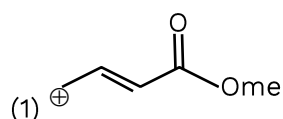
Find the major product 'A'



Ans. (1)



Q6 Stability of Carbocation is maximum in ?



Ans. (4)

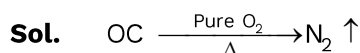


Less electronegative more electron donor

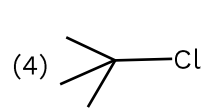
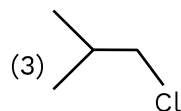
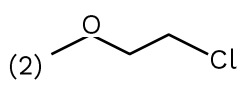
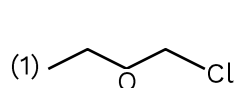
Q7 In Duma's which gas is evolved?

- (1) N_2 (2) O_2 (3) SO_2 (4) SO_3

Ans. (1)



Q8 Which of the following give nucleophile substitution reaction fastest?



Ans. (1)

Sol. Carbocation stable due to back bonding

Q9 Which of the following complex produce 2 mole of AgCl ppt the in presence of excess amount of AgNO₃ solution.

- (1) [Co(NH₃)₄Cl₂] (2) [Co(NH₃)₅Cl]Cl₂ (3) [Co(NH₃)₃Cl₃] (4) [Co(NH₃)₆]Cl₃

Ans. (2)

Sol. [Co(NH₃)₅Cl]Cl₂ $\xrightarrow{\text{AgNO}_3}$ 2 mole of AgCl

Q10 If the K_{sp} of Cr(OH)₃ is 1.6 × 10⁻¹³ M⁴. The molar solubility of salt in water is 1.56 × 10^{-x}, then value of 'x' is

Ans. (8)

Sol. $K_{sp} = 27(s)^4 = 16 \times 10^{-14} (s)^4$

$$(S)^4 = \frac{16}{27} \times 10^{-14} = \left(\frac{1600}{27} \right) \times 10^{-16}$$

$$(S) = \left(\frac{1600}{27} \right)^{1/4} \times 10^{-4}$$

$$\approx 2.77 \times 10^{-4}$$

Q11 At the freezing point of water process is non-spontaneous, at boiling point it is spontaneous (temperature varies linearly with pressure). The correct option is -

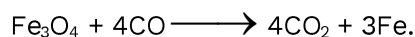
- (1) Both ΔH & ΔS are negative (2) Both ΔH & ΔS are positive
(3) ΔS negative & ΔH positive (4) ΔS positive & ΔH negative

Ans. (2)

Sol. $T \uparrow \Rightarrow \Delta G = -ve$

$$\Delta G = \Delta G = \frac{\Delta H}{+ve} - T \frac{\Delta S}{+ve}$$

Q12 If 10 mol of CO and 10 mol of Fe₃O₄ reacts according to



The weight of Fe produced in the reaction is

Ans. 420

Sol. $\text{Fe}_3\text{O}_4 + 4\text{CO} \longrightarrow 4\text{CO}_2 + 3\text{Fe}$

10 mol 10 mol



Limiting reagent

4 mol of CO → 3 mole of Fe

10 mol of CO → $\frac{3}{4} \times 10$ mole

$$\text{Weight of Fe} = \frac{3}{4} \times 10 \times 56$$

$$= 420 \text{ g}$$

Q13 In the preparation of potassium permanganate from pyrolusite ore (MnO_2). The fusion of pyrolusite ore is done with an alkali metal hydroxide like KOH in presence of air or an oxidizing agent like KNO_3 . Which first produces



Product may be -

- (1) K_2MnO_4 (2) KMnO_4 (3) Mn_3O_4 (4) K_3MnO_4

Ans. (1)

Sol. NCERT based

Q14 Ribose present in D.N.A is

- (A) It is pentose sugar
(B) Present in pyranose form
(C) α - Anomeric carbon is present
(D) It is reducing sugar in free form
(E) It is present in D configuration

Choose the correct statements.

- (1) A, C, E only (2) A, D, E only (3) A, B, C, D, E (4) A and E only

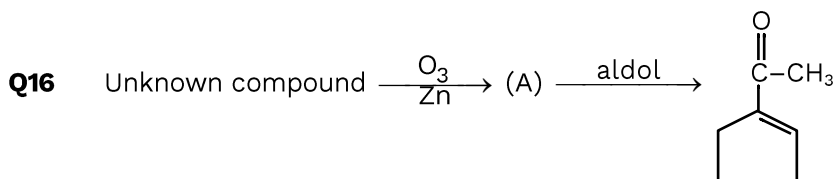
Ans. (2)

Sol. NCERT based

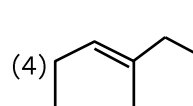
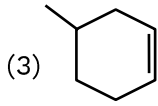
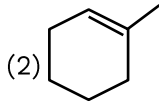
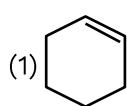
Q15 Which of the following is most reactive towards nucleophilic addition reactions

- (1) Para-nitro benzaldehyde (2) Para-methyl benzaldehyde
(3) Benzaldehyde (4) Acetophenone

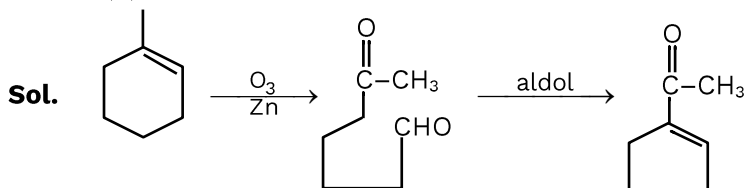
Sol. Due to electron withdrawing group $-\text{NO}_2$ in para nitrobenzaldehyde, it is more reactive.



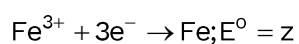
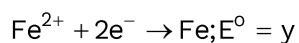
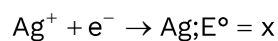
Unknown compound can be ?



Ans. (2)



Q17 $\text{Fe}^{2+} + \text{Ag}^+ \rightarrow \text{Fe}^{3+} + \text{Ag}$, E_{net}°



The value of $E_{\text{net}}^{\circ} = ?$

(1) $x + y - z$

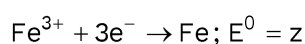
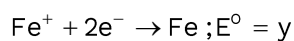
(2) $x + 3y - 2z$

(3) $y - 2x$

(4) $x - 3z + 2y$

Ans. (4)

Sol. $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}; E^{\circ} = x$



$$\Delta G_{\text{net}}^{\circ} = \Delta G_1^{\circ} + \Delta G_2^{\circ} - \Delta G_3^{\circ}$$

$$-nFE_{\text{cell}}^{\circ} = -nFE_1^{\circ} + (-nFE_2^{\circ}) - nFE_3^{\circ}$$

$$+E_{\text{cell}}^{\circ} = 1 \times x + 2 \times y - 3 \times z$$

$$E_{\text{cell}}^{\circ} = x + 2y - 3z$$

Q18 How many of the following cation shows characteristic coloured ppt. with $\text{K}_4[\text{Fe}(\text{CN})_6]$?

Cu^{2+} , Ca^{2+} , Ba^{2+} , Fe^{3+} , Zn^{2+} , Mg^{2+} , Mn^{2+}

Ans. (3)

Sol. Cu^{2+} , Fe^{3+} , Mn^{2+}

NCERT Based.